Integral Transforms of Analytic Functions of the Derivative Operator and Associated Results

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From the classical umbral-calculus relation $e^D = T$, relating the derivative operator D and the shift operator T, to binomial expansions of $(x + dy/dx)^n$, operators which arise from power expansions of the derivative are abundant.

First, we present a simple yet useful lemma which relates power series of the derivative operator to integral transforms;

$$j(D)[f] = \mathcal{L}^{-1}(j) * f$$

And explore the tight relationship between the integral transform view of differential operators and their formal or umbral analog.

1 Analytic Functions of the Derivative Operator

Taking a broader look at the family of derivative curiosities made from power series, it is notable that each are compactly represented under integral transforms. Taking Fourier as a chief example;

$$F\left\{D \cdot f(t)\right\} = \int_{-\infty}^{\infty} f'(t)\, e^{-2\pi itw} dt \to 2\pi iw \cdot F(w)$$

$$F\left\{e^D \cdot f(t)\right\} = \int_{-\infty}^{\infty} f(t+1)\, e^{-2\pi itw} dt \to e^{2\pi iw} \cdot F(w)$$

$$F\left\{D! \cdot f(t)\right\} = \int_{-\infty}^{\infty} \left[\int_0^{\infty} e^{-s} f(t + \ln(s))\, ds\right] e^{-2\pi itw} dt \to (2\pi iw)! F(w)$$

From these observations it is sensible to posit that $D \leftrightarrow 2\pi iw$ under the Fourier transform. *More generally*, if J is a derivative operator of the the following form:

$$J = j(D) = \sum_{n=0}^{\infty} a_n D^n$$

Linearity of the Fourier transform reduces everything down to;

$$F\left\{J \cdot f(t)\right\} = \sum_{n=0}^{\infty} a_n F\left(D^n \cdot f(t)\right) = \sum_{n=0}^{\infty} a_n \left(2\pi iw\right)^n F(w) = j(2\pi iw)\, F(w)$$

In particular, this permits the expression of any Taylor series-based derivative operator in convolution form;

$$J \cdot f(t) = F^{-1}\left\{j(2\pi iw)\right\} * F^{-1}\left\{F(w)\right\} = F^{-1}\left\{j(2\pi iw)\right\} * f(t) = L^{-1}(j) * f(t)$$

Where L^{-1} is the inverse Laplace transform. For example, the e^D case reduces to the delta function, under which convolution produces the shift operator. As another example, sums over derivatives can be expressed in integral form as shown,

$$\sum_{n=0}^{\infty} (-1)^n f^{(n)}(x) \to \frac{1}{1+D} \to L^{-1}\left(e^{-t}\right) \to \int_{-\infty}^{\infty} f(x-t)\left(u(t)\, e^{-t}\right) dt = \int_0^{\infty} f(x-t)\, e^{-t} dt$$

(For the unit step function $u(t)$). This fact is intriguing but nonetheless trivial to show by taking transforms. The reduction of a series of derivatives to an integral should not come as a surprise, as integration-by-parts achieves precisely this;

$$\int f(x-t)\, e^{-t} dt = -\sum_{n} (-1)^n \cdot (-1)^n f^{(n)}(x-t) \cdot (-1)^n e^{-t}$$

Similarly of course, analytic functions of the shift operator follow a similar pattern;

$$j(T) \cdot f = \sum_{n=0}^{\infty} a_n f(x+n) \to F\{j(T) \cdot f\} = \sum_{n=0}^{\infty} a_n F\{f(x+n)\} = \sum_{n=0}^{\infty} a_n e^{2\pi i w n} F(w) = j\left(e^{2\pi i w}\right) F(w)$$

Or another quick example relating to the inverse of the derivative operator, accumulation under integration;

$$\ln\left(\frac{1}{1-\frac{1}{D}}\right) f = L^{-1}\{\ln(D) - \ln(D-1)\} * f = \frac{e^t - 1}{t} * f = \int_0^x \frac{e^{(x-t)} - 1}{x-t} f(t)\, dt =$$

$$\sum_{n=1}^{\infty} \frac{1}{n} \cdot \frac{1}{(n-1)!} \int_0^x (x-t)^{(n-1)} f(t)\, dt = \sum_{n=1}^{\infty} \frac{I^n}{n} f = \ln\left(\frac{1}{1-I}\right) f$$

As a final curious application, suppose we relax the analyticity condition and investigate the modulus of the derivative;

$$|D| f \to f(t) * F^{-1}\{|2\pi i w|\} = \frac{1}{-\pi} f(t) * F^{-1}\left\{\frac{-\pi}{|2\pi w|^{(-2+1)}}\right\} =$$

$$\frac{1}{-\pi} f(t) * F^{-1}\left\{\lim_{a \to -2} \frac{-2\sin\left(\frac{\pi a}{2}\right)\Gamma(a+1)}{|2\pi w|^{(a+1)}}\right\} = \frac{1}{-\pi} f(t) * |t|^{-2} = \frac{1}{-\pi} \int_{-\infty}^{\infty} \frac{f(x-t)}{t^2} dt$$

Consider also the following analogy; applying analytic functions, denoted j, of matrix arguments is equivalent to scalar evaluation on the eigenvalues.

$$j \circ \mathrm{eig}\{A\} = \mathrm{eig}\{j \circ A\} \Rightarrow \mathcal{F}\{j \circ A\} = \mathcal{F}\{j\}\, \mathcal{F}\{A\}$$

2 An Example: Factorial Differentiation

Let D be the derivative operator and T the unit shift operator. By a formal power series it is understood that $e^D = T$. Thus the *Factorial Derivative* is defined from its parent Gamma Function integral;

$$D![f(x)] = f \int_0^{\infty} e^{-t} t^D dt = f \int_0^{\infty} e^{-t} e^{\ln(t) D} dt = \int_0^{\infty} e^{-t} f(x + \ln(t))\, dt$$

And by testing some typical functions, several peculiar patterns appear;

$$D![c] = c\ ,\ D![x] = x - \gamma\ ,\ D![x^2] = (x-\gamma)^2 + \frac{\pi^2}{6}\ ,\ D!\ [x^3] = (x-\gamma)^3 - 2\zeta(3) - \frac{\gamma\pi^2}{2} + 3x\frac{\pi^2}{6}$$

$$D!\ [x^n] = \sum_{k=0}^{n} \mathrm{nCr}(n,k)\, x^{(n-k)} \Gamma^{(k)}(1) = (x+D)^n [\Gamma(1)]$$

Interestingly, for polynomials the Factorial Derivative is no longer nilpotent in regards to reducing the degree of the input. Next;

$$D!\left[e^{bx}\right] = e^{bx} \cdot b!$$

$$D![\cos(ax)] = \cos(ax)\, \Gamma_{Real}(ia+1) - \sin(ax)\, \Gamma_{Imag}(ia+1)$$

$$D![\sin(ax)] = \sin(ax)\, \Gamma_{Real}(ia+1) + \cos(ax)\, \Gamma_{Imag}(ia+1)$$

Which are easily seen from taking imaginary and real parts from $f = e^{iax}$. Under the context of $D!$ as an integral transform we have;

$$D![af(x) + bg(x)] = aD![f(x)] + bD![g(x)]$$

3 Differential & Shift Equations

While the utility of integral transforms for differential equations are well established, problems involving difference and shift evaluations can be articled cleanly in a similar manner. Suppose, for example, we wish to solve the peculiar

$$y' = \Delta y + G$$

Under the use of integral transforms this problem becomes trivial. Using inverse Laplace:

$$-tf(t) = e^{-t} f(t) - f(t) + g(t) \to f(t) = \frac{g(t)}{1 - t - e^{-t}}$$

Thus;

$$y(s) = \int_0^\infty \frac{e^{-st} \cdot \mathcal{L}^{-1}\{G\}[t]}{1 - t - e^{-t}} dt$$

As an additional note, the equation $y' = \Delta y$ has several intriguing properties coupling the continuous and discrete calculus. A solution family;

$$y(t) = c_0 e^{(-1 - W_q(-\frac{1}{e}))t} + c_1 t + c_2$$

For the various complex-valued branches of the Lambert-W function, y has the peculiar property that;

$$\sum_{n=a}^{b-1} y'(n) = \int_a^b \Delta y(t)\, dt$$

Whilst the following stronger case holds;

$$\int_m^n ce^{x\left(-1 - W_q\left(-\frac{1}{e}\right)\right)} + c_1 dx = \sum_{k=m}^{n-1} ce^{n\left(-1 - W_q\left(-\frac{1}{e}\right)\right)} + c_1$$

A similar Fourier formulation can be reduced via the Residue Theorem;

$$f'(t) = f(t+1) - f(t) + a\delta(t) \to F(w) = \frac{1}{2\pi i w - e^{2\pi i w} + a} \to f(t) = \int_{-\infty}^\infty \frac{e^{2\pi i t w}}{2\pi i w - e^{2\pi i w} + a} dw$$

$$\Rightarrow f(t) = \frac{e^{2\pi i t \xi}}{2\pi i} \left(\frac{1}{1 - e^{2\pi i \xi}} \right)$$

Where ξ is the root of $2\pi i \xi - e^{2\pi i \xi} + a = 0$.

4 Another Example: Grünwald–Letnikov derivative

It ought to be stated more outright that integral transforms such as Fourier are precisely the machinery which allow for the umbral derivative operations which have been used throughout.

In operator fashion we would derive Grünwald–Letnikov by a limit;

$$D^q = \ln(T)^q = \lim_{h\to 0} \left(\frac{1 - T^h}{h} \right)^q = \lim_{h\to 0} \frac{1}{h^q} \sum_{m=0}^\infty \mathrm{nCr}(q,m) \left(-T^h\right)^m = \lim_{h\to 0} \frac{1}{h^q} \sum_{m=0}^\infty (-1)^m \mathrm{nCr}(q,m) f(x - hm)$$

Note the similarities in the derivation by Fourier transform;

$$D^q f = F^{-1}\{F\{f\}(2\pi i w)^q\} = \lim_{h\to 0} F^{-1} \left\{ F\{f\} \frac{\left(1 - e^{2\pi i w h}\right)^q}{h^q} \right\} = \lim_{h\to 0} F^{-1} \left\{ F\{f\} \frac{1}{h^q} \sum_{m=0}^\infty (-1)^m \mathrm{nCr}(q,m) e^{-2\pi i w h m} \right\} =$$

$$\lim_{h\to 0} F^{-1} \left\{ \frac{1}{h^q} \sum_{m=0}^\infty (-1)^m \mathrm{nCr}(q,m) F\{f(x - hm)\} \right\} = \lim_{h\to 0} \frac{1}{h^q} \sum_{m=0}^\infty (-1)^m \mathrm{nCr}(q,m) f(x - hm)$$

Each of the properties of the operator, the factorization from the function, the linearity, and combination under multiplication, are but reflections of the much more rigorous integral transform.

5 Stochastic Symbology

Integral Transforms are of course primary instruments for the solution of PDEs–let us present a convenient notion which escapes their use. Here is a simple stochastic interpretation of the solution to the diffusion equation, well known to be convolution with a Gaussian.

$$\frac{\partial u}{\partial t} = \frac{1}{2}\alpha^2 \frac{\partial^2 u}{\partial x^2} \ ; \ u(0,s) = f(s)$$

At every point along the curve of $f(x)$ there is a flux of particles, heat, or other matter which jostle and flow in a manner analogous to Brownian motion. The rate of this Brownian flow has the strength α, and as time increases will reach further and further along the spatial axis as the regions become homogenized.

Whatever the *expected* state of the system is, following these continual stochastic interactions will represent the stable curve for u, and consequently we find;

$$u = E\left\{f\left(x + \alpha B_t\right)\right\}$$

Positing the equivalence in mean and variance of $B_t \sim \sqrt{t}N\left(0,1\right)$,

$$u = \int_{-\infty}^{\infty} f\left(x + \alpha\sqrt{t}z\right)\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz = \frac{1}{\alpha\sqrt{2\pi t}}\int_{-\infty}^{\infty} f\left(y\right)e^{-\frac{(y-x)^2}{2\alpha^2 t}}dy = f * \phi\left(x, t, k = \frac{1}{2}\alpha^2\right)$$

However mundane this perspective may seem, it offers explanations for otherwise algebraic corollaries;

If f is linear, u makes no change to f over time. This simple fact is equivalent to the notion that $f\left(x + \alpha B_t\right)$, by the $\frac{\partial f}{\partial t} = -\frac{1}{2}\frac{\partial^2 f}{\partial B_t{}^2}$ condition, is a *martingale*: the Brownian exchange of diffusion possesses no drift, and;

$$u\left(t > 0\right)|\mathcal{F}_{t=0} = E\left\{f\left(x + \alpha B_t\right)|\mathcal{F}_{t=0}\right\} = E\left\{f\left(x + \alpha B_t\right)|_{t=0}\right\} = f\left(x\right)$$

Physical processes are of course 'updated' by all prior states.

6 Miscellaneous Operators

Further corollaries, discussions, and related activities include:

1. Referring to the standard theory of the Transport PDE and the Heat PDE, if $\exp\left(tD\right)$ produces a shift operator and if $\exp\left(tD^2\right)$ produces a diffusion operator, what physical 'movement' interpretation is assigned to the general $\exp\left(tD^k\right)$?

2. What is furthermore interesting about the $D \leftrightarrow 2\pi i w$ relationship from the Fourier formalization is that the act of 'dividing by zero' in differentiation takes a suspiciously similar form;

$$\frac{1}{0} \to \frac{1}{\ln\left(1\right)} \to \frac{1}{\ln\left(e^{2\pi i k}\right)} \to \frac{1}{2\pi i k}$$

3. *Functions of Accumulation:* From the perspective of the derivative operator, integration (from a to x) is;

$$\frac{e^{Da} - 1}{D} f\left(x\right) \ ;$$

   Further, while the exponent of D is the translation operator T, the exponent of accumulation is (for modified Bessel function I_0).

$$e^{z\int_a^x} f\left(x\right) = \int_0^{2\sqrt{(x-a)z}} I\left(s\right) f\left(x - \frac{s^2}{4z}\right) ds$$

   And continuing the exponentiation trend, should one exponentiate a reflection matrix, rather than a rotation matrix $\sqrt{-1}$, one will receive terms collapsing to cosh and sinh.

4. *Adjoint Operators:* Integration by Parts does not escape an umbral generalization to discrete and integral-transform settings, giving suspiciously similar identities such as the following.

   For derivatives we have;

$$\int f \cdot D\,[g] \;=\; fg - \int D\,[f]\,g$$

   For discrete shifts we have; (*Summation by Parts*)

$$\sum f \cdot \Delta g = fg - \sum g \cdot \Delta f$$

   For Laplace and Fourier Transforms also;

$$\int_{\mathcal{R}} L\,\{f\}\,g = \int_{\mathcal{R}} f L\,\{g\}$$